Package 'mlsas'

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Type Package

Title A Machine-Learning-Based Spatial Autocorrelation Statistic

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Description Learns attribute similarity with an unsupervised random forest and computes global and local spatial autocorrelation statistics. Provides total and conditional Monte Carlo randomization inference, spatial weights, small synthetic examples, and explicit input validation. Implements the method of Kim and Lee (2026) <doi:10.1111/gean.70062>.

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Imports randomForest, Matrix, sf, spdep, parallel, stats, graphics, utils

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URL https://github.com/geowhk/ml-sas-r, https://doi.org/10.1111/gean.70062

BugReports https://github.com/geowhk/ml-sas-r/issues

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mlsas-package *A Machine-Learning-Based Spatial Autocorrelation Statistic*

Description

Implementation of learned attribute similarity and spatial randomization inference.

Details

Start with `mlsas_example` and `mlsas`. Default settings follow the empirical analysis in the accompanying paper. The tutorial provides complete small-data workflows.

Author(s)

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References

Kim, W. and Lee, S.-I. (2026). ML-SAS: A Machine-Learning-Based Spatial Autocorrelation Statistic for Mixed-Type Data. *Geographical Analysis*, 58(4), e70062. doi:10.1111/gean.70062.

mlsas *Run an ML-SAS Analysis*

Description

Fit attribute similarity and perform global and local spatial autocorrelation tests. Default settings follow the empirical analysis in the accompanying paper.

Usage

```
mlsas(data, variables, id, spatial = NULL, weights = NULL,
    contiguity = "queen", style = "W", proximity = "path", ntree = 1000L,
    mtry = NULL, nodesize = 1L, nperm = 99999L, randomization = "conditional",
    alternative = "two.sided", p_adjust = "BH", alpha = 0.05,
    seed = NULL, workers = 1L, verbose = TRUE, progress = interactive())
```

Arguments

`data`	An sf object or a data.frame/tibble containing attributes and the ID column.
`variables`	Character vector selecting analysis columns explicitly. Exclude IDs and geometry. Numeric, factor and ordered-factor attributes are supported; no automatic encoding or imputation.
`id`	Name of a unique, non-missing region ID column. Required for the integrated function. In the weights constructor, omitted IDs use row names.
`spatial`	An sf object. Separate attribute and spatial tables must have identical ID sets; spatial rows are reordered to attribute order. The weights constructor also accepts a square, ID-labelled numeric matrix.

`weights`	Square non-negative weights matrix with matching row and column IDs. Supplied weights are used as provided, without implicit standardization. Construct transformed weights with mlsas_weights first.
`contiguity`	Polygon contiguity: queen includes shared vertices; rook requires a shared boundary segment.
`style`	Weight transform: W row-standardizes, B makes positive edges binary, raw preserves supplied magnitudes. Applies when constructing weights, not to an existing matrix supplied to mlsas.
`proximity`	Path uses the mean full-root-to-terminal path Jaccard similarity; terminal uses mean leaf co-membership.
`ntree`	Positive integer number of trees. Default 1000 follows the empirical analysis.
`mtry`	Number of candidate attributes per split. NULL uses max(1, floor(sqrt(number of attributes))), the classification forest rule used by the empirical code.
`nodesize`	Positive integer minimum terminal node size; default 1 follows the empirical classification forest.
`nperm`	Positive integer Monte Carlo draws, default 99999. Explicit permutations override this value. With one draw, permutation standard deviations are NA.
`randomization`	conditional fixes the focal profile; total permutes all profile labels jointly. Local default is conditional.
`alternative`	two.sided (default), greater, or less. Two-sided p is min(1, 2 * min(p_high, p_low)).
`p_adjust`	A method accepted by stats::p.adjust, default BH. Applied across all input regions within this local test; not across separate models.
`alpha`	Significance threshold strictly between zero and one, default 0.05.
`seed`	NULL consumes the current R random-number stream. An integer sets a local reproducible stream and restores the caller RNG state afterwards. The integrated function uses one sequential stream for fitting and both tests.
`workers`	Positive number of proximity-calculation workers, default 1. Forest fitting and permutation tests are serial in version 0.1.0. Fixed tree groups retain summation order across worker counts.
`verbose`	Logical; show start, stage and elapsed-time messages. Does not request interactive confirmation.
`progress`	Logical; show actual completed tree/permutation counts. Default interactive(). Forest fitting reports running because the engine exposes no progress callback.

Details

Self-similarity and weight diagonals are excluded. With n regions, $P = n(n-1)W/\sum W$ and $Q = S/\sum S$. The global statistic is $\sum_{ij} P_{ij}Q_{ij}$ and the local statistic is $n\sum_j P_{ij}Q_{ij}$. The local average equals global ML-SAS. The fitted representation is held fixed during inference. Tail counts include ties; Monte Carlo p-values use (count + 1)/(B + 1). Significant locations above their permutation mean are similarity clusters; those below it are dissimilarity/outliers. These are not univariate High-High/Low-Low labels. An equality to the permutation mean is not assigned a directional discovery. All-tied distributions yield p = 1. No observations are silently removed. Dense similarity storage is quadratic in n. Inference streams batches of 256 draws. User-supplied permutation indices use additional n by B storage.

Value

An mlsas_fit list containing statistic, global, local, similarity, spatial and settings.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
fit <- mlsas(d$data, d$variables, d$id, ntree = 8, nperm = 19,
             seed = 7, verbose = FALSE, progress = FALSE)
summary(fit)
```

---

mlsas-methods *Inspect ML-SAS Results*

Description

Display observed statistics, test tables, effective settings, and maps or index plots.

Usage

```
## S3 method for class 'mlsas_fit'
print(x, ...)
## S3 method for class 'mlsas_stat'
print(x, ...)
## S3 method for class 'mlsas_test'
print(x, ...)
## S3 method for class 'mlsas_fit'
summary(object, ...)
## S3 method for class 'mlsas_test'
summary(object, ...)
## S3 method for class 'mlsas_test'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
## S3 method for class 'mlsas_fit'
plot(x, ...)
## S3 method for class 'mlsas_test'
plot(x, ...)
```

Arguments

x, object	An ML-SAS result of the corresponding class.
row.names, optional	Passed to the data.frame conversion.
...	Additional printing or plotting arguments; unused by summary methods.

Details

A fitted object with geometry plots local classes on that geometry. Without geometry it plots local statistics against region index. Test plots also use region index.

Value

Print and plot return the input invisibly. Summary returns tables and settings; as.data.frame returns the test table.

mlsas_example *Small Synthetic Spatial Examples*

Description

Return a deterministic 6 by 6 square lattice with continuous and categorical attributes for learning the API.

Usage

```
mlsas_example(type = c("mixed", "continuous"))
```

Arguments

type	mixed (numeric, nominal and ordered attributes) or continuous. Both return 36 synthetic lattice regions.

Value

A list containing data (sf), attributes, spatial, x, weights, variables and id.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
names(d)
```

mlsas_global_test *Global Randomization Inference*

Description

Test global association by simultaneous row and column label permutations of the learned similarity matrix.

Usage

```
mlsas_global_test(object, alternative = "two.sided", nperm = 99999L,
    seed = NULL, alpha = 0.05, keep = FALSE, verbose = TRUE,
    progress = interactive(), permutations = NULL)
```

Arguments

object	An mlsas_stat object returned by mlsas_stat().
alternative	two.sided (default), greater, or less. Two-sided p is min(1, 2 * min(p_high, p_low)).
nperm	Positive integer Monte Carlo draws, default 99999. Explicit permutations override this value. With one draw, permutation standard deviations are NA.
seed	NULL consumes the current R random-number stream. An integer sets a local reproducible stream and restores the caller RNG state afterwards. The integrated function uses one sequential stream for fitting and both tests.
alpha	Significance threshold strictly between zero and one, default 0.05.
keep	Logical; retain permutation statistics (regions by draws for a local test). Default FALSE avoids storing the full distribution.
verbose	Logical; show start, stage and elapsed-time messages. Does not request interactive confirmation.
progress	Logical; show actual completed tree/permutation counts. Default interactive(). Forest fitting reports running because the engine exposes no progress callback.
permutations	Optional n by B matrix, each column a permutation of integers 1:n. Intended for paired tests and exact-enumeration checks. Even with exhaustive inputs, the documented Monte Carlo +1 p-value convention is retained.

Details

Self-similarity and weight diagonals are excluded. With n regions, $P = n(n-1)W/\sum W$ and $Q = S/\sum S$. The global statistic is $\sum_{ij} P_{ij}Q_{ij}$ and the local statistic is $n\sum_j P_{ij}Q_{ij}$. The local average equals global ML-SAS. The fitted representation is held fixed during inference. Tail counts include ties; Monte Carlo p-values use (count + 1)/(B + 1). Significant locations above their permutation mean are similarity clusters; those below it are dissimilarity/outliers. These are not univariate High-High/Low-Low labels. An equality to the permutation mean is not assigned a directional discovery. All-tied distributions yield p = 1. No observations are silently removed. Dense similarity storage is quadratic in n. Inference streams batches of 256 draws. User-supplied permutation indices use additional n by B storage.

Value

An mlsas_test list with table, settings and optional permutation statistics.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
s <- mlsas_similarity(d$x, ntree = 8, seed = 7,
                      verbose = FALSE, progress = FALSE)
o <- mlsas_stat(s, d$weights)
test <- mlsas_global_test(o, nperm = 19, seed = 8,
                          verbose = FALSE, progress = FALSE)
as.data.frame(test)
```

`mlsas_local_test` *Local Randomization Inference*

Description

Test each region using conditional or total label randomization and adjust local p-values.

Usage

```
mlsas_local_test(object, randomization = "conditional", alternative = "two.sided",
    nperm = 99999L, seed = NULL, alpha = 0.05, p_adjust = "BH",
    keep = FALSE, verbose = TRUE, progress = interactive(), permutations = NULL)
```

Arguments

`object`	An mlsas_stat object returned by mlsas_stat().
`randomization`	conditional fixes the focal profile; total permutes all profile labels jointly. Local default is conditional.
`alternative`	two.sided (default), greater, or less. Two-sided p is min(1, 2 * min(p_high, p_low)).
`nperm`	Positive integer Monte Carlo draws, default 99999. Explicit permutations override this value. With one draw, permutation standard deviations are NA.
`seed`	NULL consumes the current R random-number stream. An integer sets a local reproducible stream and restores the caller RNG state afterwards. The integrated function uses one sequential stream for fitting and both tests.
`alpha`	Significance threshold strictly between zero and one, default 0.05.
`p_adjust`	A method accepted by stats::p.adjust, default BH. Applied across all input regions within this local test; not across separate models.
`keep`	Logical; retain permutation statistics (regions by draws for a local test). Default FALSE avoids storing the full distribution.
`verbose`	Logical; show start, stage and elapsed-time messages. Does not request interactive confirmation.
`progress`	Logical; show actual completed tree/permutation counts. Default interactive(). Forest fitting reports running because the engine exposes no progress callback.
`permutations`	Optional n by B matrix, each column a permutation of integers 1:n. Intended for paired tests and exact-enumeration checks. Even with exhaustive inputs, the documented Monte Carlo +1 p-value convention is retained.

Details

Self-similarity and weight diagonals are excluded. With n regions, $P = n(n-1)W/\sum W$ and $Q = S/\sum S$. The global statistic is $\sum_{ij} P_{ij}Q_{ij}$ and the local statistic is $n\sum_j P_{ij}Q_{ij}$. The local average equals global ML-SAS. The fitted representation is held fixed during inference. Tail counts include ties; Monte Carlo p-values use (count + 1)/(B + 1). Significant locations above their permutation mean are similarity clusters; those below it are dissimilarity/outliers. These are not univariate High-High/Low-Low labels. An equality to the permutation mean is not assigned a directional discovery. All-tied distributions yield p = 1. No observations are silently removed. Dense similarity storage is quadratic in n. Inference streams batches of 256 draws. User-supplied permutation indices use additional n by B storage.

Value

An mlsas_test list with table, settings and optional permutation statistics.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
s <- mlsas_similarity(d$x, ntree = 8, seed = 7,
                      verbose = FALSE, progress = FALSE)
o <- mlsas_stat(s, d$weights)
test <- mlsas_local_test(o, nperm = 19, seed = 8,
                         verbose = FALSE, progress = FALSE)
as.data.frame(test)
```

---

mlsas_similarity *Learn Attribute Similarity*

Description

Fit an Addcl1 real-versus-synthetic classification forest and average path or terminal similarities.

Usage

```
mlsas_similarity(x, proximity = c("path", "terminal"), ntree = 1000L,
    mtry = NULL, nodesize = 1L, seed = NULL, workers = 1L, verbose = TRUE,
    progress = interactive(), synthetic = NULL)
```

Arguments

x	Attribute data.frame/tibble without geometry, with region IDs as row names. For S3 methods, a package result object.
proximity	Path uses the mean full-root-to-terminal path Jaccard similarity; terminal uses mean leaf co-membership.
ntree	Positive integer number of trees. Default 1000 follows the empirical analysis.
mtry	Number of candidate attributes per split. NULL uses max(1, floor(sqrt(number of attributes))), the classification forest rule used by the empirical code.
nodesize	Positive integer minimum terminal node size; default 1 follows the empirical classification forest.
seed	NULL consumes the current R random-number stream. An integer sets a local reproducible stream and restores the caller RNG state afterwards. The integrated function uses one sequential stream for fitting and both tests.
workers	Positive number of proximity-calculation workers, default 1. Forest fitting and permutation tests are serial in version 0.1.0. Fixed tree groups retain summation order across worker counts.
verbose	Logical; show start, stage and elapsed-time messages. Does not request interactive confirmation.

progress	Logical; show actual completed tree/permutation counts. Default interactive(). Forest fitting reports running because the engine exposes no progress callback.
synthetic	Optional Addcl1 sample for paired checks: same dimensions, names, column classes and factor levels as x. NULL independently resamples each attribute with replacement.

Details

Numeric, nominal and ordered factors are supported. Missing values, constant attributes and nominal factors with more than 31 levels are rejected in this release. Ordered factors follow their declared order. The nominal-level limit protects the inherited integer bit-mask path decoder. No factor is silently one-hot encoded. Progress reporting does not alter forest fitting. The full similarity matrix requires quadratic memory in the number of regions.

Value

A symmetric numeric similarity matrix, with region-ID dimnames and effective settings. Its diagonal is one; downstream statistics exclude the diagonal.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
s <- mlsas_similarity(d$x, ntree = 8, seed = 7,
                      verbose = FALSE, progress = FALSE)
s[1:3, 1:3]
```

---

mlsas_stat *Compute Global and Local ML-SAS*

Description

Calculate observed statistics from a similarity matrix and spatial weights.

Usage

```
mlsas_stat(similarity, weights)
```

Arguments

similarity	Symmetric finite non-negative matrix with matching row/column region IDs. Its diagonal is excluded; positive off-diagonal sum is required. User-supplied similarities define the attribute representation used in the analysis.
weights	Square non-negative weights matrix with matching row and column IDs. Supplied weights are used as provided, without implicit standardization. Construct transformed weights with mlsas_weights first.

Details

Self-similarity and weight diagonals are excluded. With n regions, $P = n(n-1)W/\sum W$ and $Q = S/\sum S$. The global statistic is $\sum_{ij} P_{ij}Q_{ij}$ and the local statistic is $n\sum_j P_{ij}Q_{ij}$. The local average equals global ML-SAS. The fitted representation is held fixed during inference. Tail counts include ties; Monte Carlo p-values use (count + 1)/(B + 1). Significant locations above their permutation mean are similarity clusters; those below it are dissimilarity/outliers. These are not univariate High-High/Low-Low labels. An equality to the permutation mean is not assigned a directional discovery. All-tied distributions yield p = 1. No observations are silently removed. Dense similarity storage is quadratic in n. Inference streams batches of 256 draws. User-supplied permutation indices use additional n by B storage.

Value

An mlsas_stat object with global, local, normalized P and Q, region IDs and settings.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
s <- mlsas_similarity(d$x, ntree = 8, seed = 7,
                      verbose = FALSE, progress = FALSE)
o <- mlsas_stat(s, d$weights)
o
```

`mlsas_weights` *Construct Spatial Weights*

Description

Construct contiguity, distance or nearest-neighbour weights, or transform an existing matrix.

Usage

```
mlsas_weights(spatial, id = NULL, method = c("contiguity", "distance",
    "knn"), contiguity = c("queen", "rook"), style = c("W", "B",
    "raw"), distance = NULL, k = 4L, symmetrize = c("union",
    "mutual", "none"))
```

Arguments

`spatial`	An sf object. Separate attribute and spatial tables must have identical ID sets; spatial rows are reordered to attribute order. The weights constructor also accepts a square, ID-labelled numeric matrix.
`id`	Name of a unique, non-missing region ID column. Required for the integrated function. In the weights constructor, omitted IDs use row names.
`method`	Neighbour construction: contiguity, distance or knn. Distance and knn require POINT geometry with known CRS; choose polygon representative points explicitly.

contiguity	Polygon contiguity: queen includes shared vertices; rook requires a shared boundary segment.
style	Weight transform: W row-standardizes, B makes positive edges binary, raw preserves supplied magnitudes. Applies when constructing weights, not to an existing matrix supplied to mlsas.
distance	Positive threshold in the units returned by sf::st_distance; metres for longitude/latitude. Required with method = distance.
k	Positive integer smaller than the region count for nearest neighbours, default 4. Equal distances are broken by region ID.
symmetrize	For knn: union, mutual or none. Directed weights are retained with none. Isolated regions always produce an error.

Details

Queen/rook requires valid nonempty polygons. Point distances use the supplied CRS. Isolated regions are rejected with their IDs. No automatic CRS transformation, imputation, geometry repair or region deletion occurs. For advanced distance/knn analyses, construct weights first and supply an attribute table plus weights to mlsas.

Value

A numeric matrix with region-ID dimnames and a settings attribute.

See Also

mlsas_example, mlsas

Examples

```
d <- mlsas_example()
w <- mlsas_weights(d$data, id = "region_id", contiguity = "rook")
head(rowSums(w))
```

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